Chapter 9
Style Production Planning Overview

Introduction

Style Production Planning is made up of the following two modules:

P2	Style Master Production Scheduling
P3	Style Material Requirements Planning

MRP and MPS are autonomous applications. You may use either application stand alone. However, MRP is normally driven by MPS demand. You can use a separate MRP demand to enhance the requirements for dependent MRP items.

If you set all your styles and materials to be MPS controlled, then you do not need to run MRP. Likewise, if you set all your styles and materials to be MRP controlled, you only need to run MRP. However, MRP does not carry out capacity planning.

If you are a distributor who does not manufacture styles, then you can install either MPS or MRP. However, If you wish to run capacity planning on your purchases, use MPS rather than MRP.

Tip: The advantages of running both MPS and MRP are that you can run MPS, then stop and manipulate the data, before running MRP.

Planning Models

Master Production Scheduling and Material Requirements Planning utilise planning models.

Planning models define:

- Stockrooms. The planning process considers only materials stocked within, or finished styles received into, stockrooms in the planning model.
- Calendar. This is the production calendar for scheduling work and defining forecasting week numbers.
- Reporting profile. This comprises reporting periods for bucketing planning information on reports and enquiries and MPS/MRP reviews. A reporting profile may be based on a non-linear time horizon. For example, a three month plan might be structured as follows:

Week 1	Daily Detail
Week 2-4	Weekly detail
Week 5-8	Fortnightly detail
Week 8-12	Monthly detail

Use planning models to view the results of alternative plans. You can perform planning runs on different areas of your data and analyse the results of different runs concurrently.

Note: If you intend to run MRP using the Include Suggested MPS parameter, then you must define identical MPS and MRP models using the same codes, stockrooms and reporting profiles.

The Live Model

You can define three types of model: centralised, multi-plant and plant. Plants may be defined as live or not live.

To operate MPS or MRP, you must define at least one planning model as the live model in the Company Profile. From the live model, you can change or confirm resulting plans and schedules.

Typically, in a non multi-plant environment, you would have one live planning model. That model, that you would use in regular planning runs to manage your on-going production, would describe the current on-going situation. To simulate other plans, whilst not interfering with the day-to-day operation of your live plan, you could have other planning models.

You can store the output from any MPS run as a revision level. You can change the contents of any revision level and use any revision to generate planning information.

Master Production Scheduling

The following list shows those processes that are specific to MPS:

- Capacity planning.
- Selection of MPS styles or materials, where you have set the Planning Type to 1 on the Production Details Screen under the Style/Material Details option.
- You can amend suggested production orders, prior to confirmation. For example, you can change quantities, routes, due dates and planning filters.
- You can define a cut-off low level code, below which the run will not look for any occurrence of MPS items.
- You can use revision level codes to save different versions of the master production schedule.
- You can review demand part way through the run, allowing you to effect changes to be incorporated when the run commences.

MPS produces a high level production plan for manufactured items. The system then matches the projected demand for designated products to supply capability, based on the planning models and planning horizons that you define.

Demand is a function of sales orders, forecast, and/or manual entries or a user-defined combination. The ability to supply is a function of the availability of associated materials and resources, as defined within Style Production Definition Management.

Where necessary, the resulting plan suggests amendments to current production orders and/or suggests generating new production and purchase orders. The calculation process for the respective plan generations is simply:

Demand less Availability = Requirement

You can generate a demand forecast by using a simple annual forecast defined within MPS.

MPS is a time based plan of suggestions to meet customer due dates. The plan is derived by the system by working backwards from the customer due date specified, taking account of product lead times, to suggest a latest start date that will achieve completion by the due date.

Production orders may be individually confirmed or confirmed ‘en masse’ and purchase orders can be automatically passed through to the Requisitioning application for subsequent confirmation and transfer to the Purchase Management application in advance of further MRP planning, if required.

Capacity Planning

Capacity planning is a unique function within Master Production Scheduling, enabling you to compare available machine and labour capacity against projected work loads.

You can use either the standard planning route/BOMs or specially derived capacity routes.

Through a Capacity Review option, you can change the work load schedule at production order level.

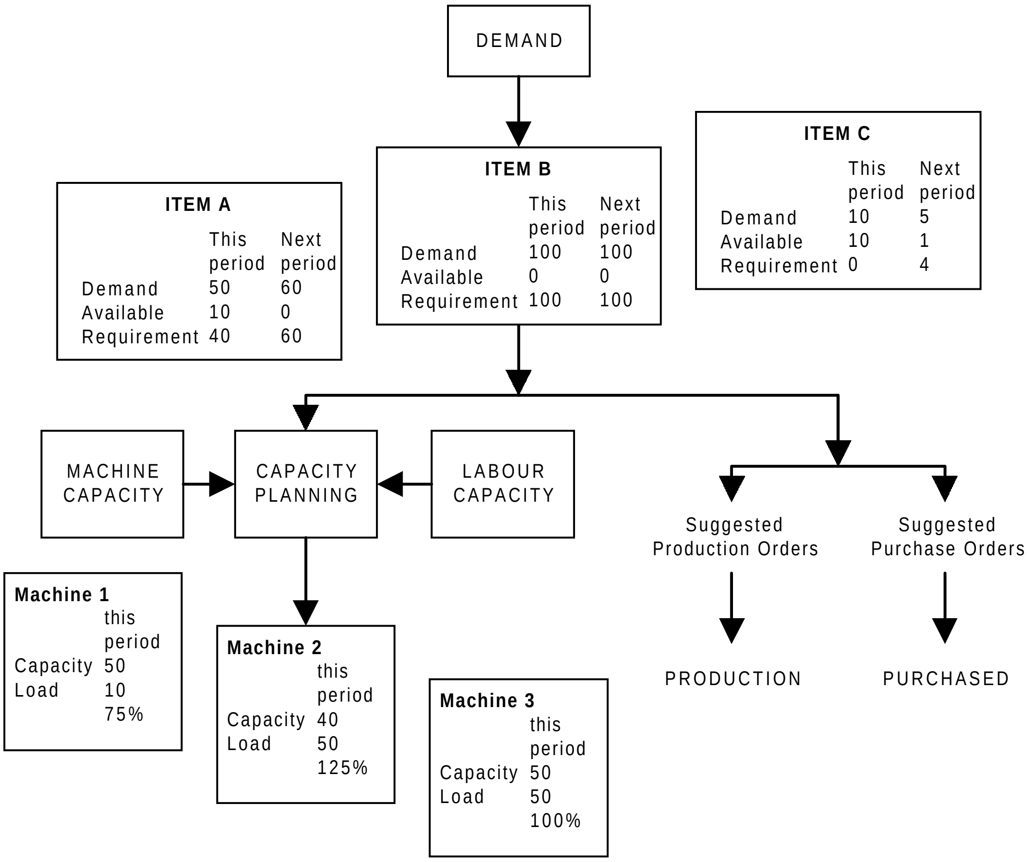

Figure : Master Production Scheduling

MPS incorporates capacity planning to:

- Take account of machine capacities when assessing the demand proposed by MPS plans.
- Determine a loading factor for each machine.
- Determine labour loading.

MPS may use summary routings to determine loading on Style Production resources.

Capacity planning goes through the following process:

1. Uses planning route information to convert demand, in quantities, into the number of hours required at each machine.
2. Uses planned supply dates to schedule those required hours into appropriate weekly production time slots at each machine. Takes the following into account: machine standard capacities; planned down times; non working days.
3. Makes available reports and enquiries from which you can compare the weekly machine/work centre hours required and available within the planning run timescales.
4. From these reports and enquiries, you can search for overload and underload at machines and for labour skills. Then, decide on any action required.

Capacity planning enables finite capacity planning by individual order, but assumes infinite capacity of the facility. It is therefore the responsibility of the planner to complete any fine-tuning of the work load to optimise the loading factor and hence utilisation of production capacity.

Material Requirements Planning

MRP produces a detailed plan for the purchase of materials.

The following list shows those processes that are specific to MRP:

- Selection of MRP styles or materials, where you have set the Planning Type to 0 on the Production Details Screen under the Style/Material Details option.
- Directly input MPS suggestions into to the material plan.

There is a choice of modes of processing:

- Regenerative. Plan for all materials.
- Selective. Plan for a range of materials rather than all the materials in the planning model.
- Net Change. Consider only those materials that have had any changes to their demand or supply since the last MRP run.

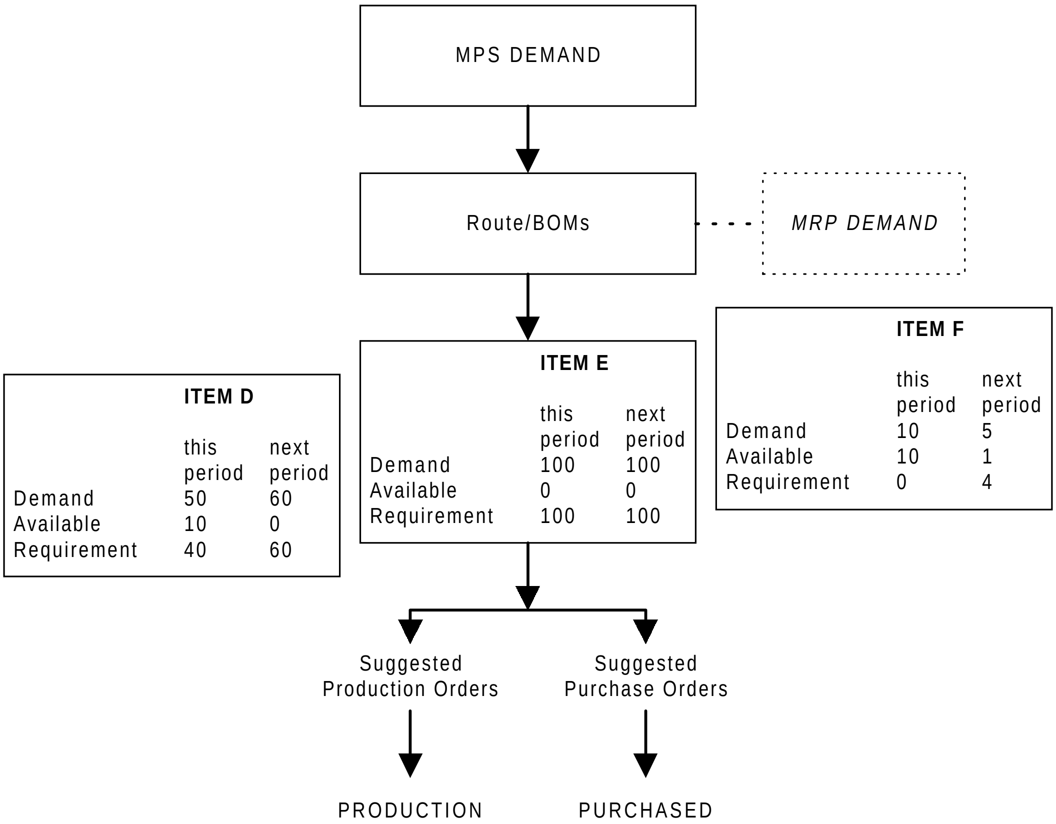

Figure : Material Requirements Planning